

Project

The alternating bit protocol

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1 Gradient Descent Algorithm

Gradient Descent Algorithm to find the minimize value of the function $f(x)$:

$$x_{i+1} = x_i - lr \times f'(x)$$

In this labwork, the funtion is $f(x) = x^2$, the initial value $x_0 = 8$ and the algorithm is running for 10 times with $lr = 0.1$:

Time	x	$f(x)$
1	8.00	64.00
2	6.40	40.96
3	5.12	26.21
4	4.10	16.78
5	3.28	10.74
6	2.62	6.87
7	2.10	4.40
8	1.68	2.81
9	1.34	1.80
10	1.07	1.15